Star-products:

Definitions and simple examples

The program known as "Deformation Quantisation" was initiated in the seminal papers [1–3] by Bayen, Flato, Frønsdal, Lichnerowicz and Sternheimer, as an alternative approach to Quantum Mechanics. For reviews, see e.g. [4–8] and references therein.

Let us start by defining what we are after, namely a star-product on a(ny) Poisson manifold.

Definition 1 [**Star-product**]**.** Let $(M, \{-,-\})$ be a Poisson manifold. A star-product on M is a map

$$\star : \mathcal{C}^\infty(M)[\![\hbar]\!] \times \mathcal{C}^\infty(M)[\![\hbar]\!] \longrightarrow \mathcal{C}^\infty(M)[\![\hbar]\!] \tag{1}$$

which is

(*i*) Unital, i.e.

$$\mathbf{1} \star f = f = f \star \mathbf{1}\,, \qquad f \in \mathcal{C}^\infty(M)\,, \tag{2}$$

where $\mathbf{1}$ denotes the unit function;

(*ii*) Associative, i.e.

$$(f \star g) \star h = f \star (g \star h)\,, \tag{3}$$

for any functions $f, g, h \in \mathcal{C}^\infty(M)$;

(*iii*) Differential, i.e.

$$f \star g = f \cdot g + \sum_{k\geq 1} \hbar^k\, B_k(f,g)\,, \tag{4}$$

where $\{B_k\}_{k\geq 1}$ are bidifferential operators, such that

$$B_1(f,g) - B_1(g,f) = 2\,\{f,g\}\,, \tag{5}$$

i.e. the antisymmetric part of the first order (in $\hbar$) bidifferential operator B_1 is the Poisson bracket of M, and $\cdot$ denotes the pointwise product of functions.

Note that this last condition implies that $\star$ is $\mathbb{R}$-bilinear.

As we will shortly see, the most important and constraining condition in the above definition is the *associativity* requirement. The condition that a star-product is unital implies that the bidifferential operators B_k vanish on constant, meaning

$$B_k(\mathbf{1}, f) = 0 = B_k(f, \mathbf{1})\,, \qquad f \in \mathcal{C}^\infty(M)\,, \tag{6}$$

for all $k \geq 1$. In plain words, this means that the bidifferential operator B_k should not contain any constant term, or equivalently, acts with at least one derivative on each argument.

On top of the above three defining properties of a star-product, here is a non-exhaustive list of additional features that a star-product may verified (or be required to verify).

- A star-product is called *natural* if the bidifferential operators B_k are of order $\leq k$ in derivatives, for all $k \geq 1$;
- A star-product is called of *Weyl-type* if the bidifferential operators B_k satisfy $B_k(f,g) = (-1)^k\, B_k(g,f)$ for all $k \geq 1$;
- A star-product is *Hermitian* if, for $\hbar$ purely imaginary (meaning upon replacing $\hbar$ with $i\,\varepsilon$ where ε is another formal parameter) the star-product verifies $\overline{f \star g} = \bar{g} \star \bar{f}$ where the bar denotes complex conjugation;
- A star-product is *G-equivariant* if, assuming that the manifold M is acted upon by a Lie group G via diffeomorphisms $R_\gamma : M \to M$ for any $\gamma \in G$, it verifies $(R_\gamma^* f) \star (R_\gamma^* g) = R_\gamma^*(f \star g)$.

With this definition in mind, let us discuss a few simple examples of star-products.

The Moyal–Weyl star-product. The simplest, and most famous example of a star-product is the Moyal–Weyl product, also known as the Moyal–Groenewold product due to its appearance in [9] and [10], which defines a quantisation of $(\mathbb{R}^{2n}, \omega)$, with

$$\omega = \omega_{ab}\, \mathrm{d}y^a \wedge \mathrm{d}y^b \,, \tag{7}$$

where the components ω_{ab} are constant. Denoting by ω^{ab} the components of the inverse of ω, with the convention that $\omega^{ac}\,\omega_{bc} = \delta^a_b$, the corresponding Moyal–Weyl star-product reads

$$(f \star_{\mathsf{MW}} g)(y) = \exp\left(\hbar\, \omega^{ab}\, \tfrac{\partial}{\partial y_1^a}\, \tfrac{\partial}{\partial y_2^b}\right) f(y_1)\, g(y_2)|_{y_1=y=y_2} \,, \tag{8}$$

where the exponential should be understood as a formal power series in $\hbar$, whose coefficients are given by the bidifferential operators obtained by repeated action of $\omega^{ab}\, \frac{\partial}{\partial y^a} \otimes \frac{\partial}{\partial y^b}$. Said differently, the above formula reads

$$f \star_{\mathsf{MW}} g = \sum_{n=0}^{\infty} \tfrac{\hbar^n}{n!}\, \omega^{a_1 b_1} \dots \omega^{a_n b_n}\, \partial_{a_1} \dots \partial_{a_n} f\, \partial_{b_1} \dots \partial_{b_n} g \,, \tag{9}$$

which makes it clear that the bidifferential operators B_k are of order k in derivatives, and obey $B_k(f,g) = (-1)^k\, B_k(g,f)$. In other words, the Moyal–Weyl star-product is an example of a *natural* star-product of *Weyl-type*.

Now let us check its associativity. To do so, it will be useful to introduce the shorthand notation

$$\pi_{ij} := \omega^{ab}\, \tfrac{\partial}{\partial y_i^a}\, \tfrac{\partial}{\partial y_j^b} = -\pi_{ji} \,, \tag{10}$$

so that the Moyal–Weyl star-product reads

$$(f \star_{\mathsf{MW}} g)(y) = \exp(\hbar\, \pi_{12}) f(y_1) g(y_2)|_{y_1=y=y_2} \,. \tag{11}$$

The star-product of three functions f, g and h reads, with this notation,

$$(f \star_{\mathsf{MW}} g) * h = \exp\big(\hbar \pi_{12}\big)(f \star_{\mathsf{MW}} g)(y_1) h(y_2)|_{y_1=y=y_2} \,. \tag{12}$$

Now since the star-product $f \star_{\mathsf{MW}} g$ is a formal power series of products (of derivatives) of f and g, each derivative with respect to y_1 acting on it will split into a sum of a derivative acting on f and g, i.e.

$$\begin{aligned} \partial_{y^a}(f \star_{\mathsf{MW}} g)(y) &= \partial_{y^a}\big[e^{\hbar\pi_{12}} f(y_1+y) g(y_2+y)\big]|_{y_1=0=y_2} && (13)\\ &= \big[(\partial_{y_1^a} + \partial_{y_2^a}) e^{\hbar\pi_{12}} f(y_1+y) g(y_2+y)\big]_{y_1=0=y_2} \,, && (14) \end{aligned}$$

by virtue of the Leibniz rule. Consequently, we can write,

$$\big[(f \star_{\mathsf{MW}} g) \star_{\mathsf{MW}} h\big](y) = \exp(\hbar\pi_{13} + \hbar\pi_{23}) \exp(\hbar\pi_{12})\, f(y_1) g(y_2) h(y_3)|_{y_1=y_2=y_3=y} \,, \tag{15}$$

and because the partial derivatives commute with one another, we can gather the exponentials to get

$$\big[(f \star_{\mathsf{MW}} g) \star_{\mathsf{MW}} h\big](y) = e^{\hbar[\pi_{12}+\pi_{13}+\pi_{23}]}\, f(y_1) g(y_2) h(y_3)|_{y_1=y_2=y_3=y} \,. \tag{16}$$

This is manifestly symmetric in the three functions f, g and h. At this point it becomes clear that we can run the previous steps backwards, and prove that

$$(f \star_{\mathsf{MW}} g) \star_{\mathsf{MW}} h = f \star_{\mathsf{MW}} (g \star_{\mathsf{MW}} h) \,, \tag{17}$$

i.e. that the Moyal–Weyl star-product is indeed associative. In fact, one can easily deduce from the previous computation a concise formula for the star-product of n functions, namely

$$(f_1 \star_{\mathsf{MW}} \dots \star_{\mathsf{MW}} f_n)(y) = \exp\Big(\hbar \sum_{1 \le i < j \le n} \pi_{ij}\Big)\, f_1(y_1) \dots f_n(y_n)|_{y_i=y} \,, \tag{18}$$

for any $f_1, \dots, f_n \in \mathcal{C}^\infty(M)$. In plain words, the Moyal–Weyl star-product of n functions is a formal power series in $\hbar$ whose coefficients are given by derivatives of all functions, contracted between all possible pairs of these functions.

Another interesting property of the Moyal–Weyl star-product is that it is equivariant with respect to the inhomogeneous symplectic group $ISp(2n,\mathbb{R}) \equiv Sp(2n,\mathbb{R}) \ltimes \mathbb{R}^{2n}$. Recall that this group consists of pairs (M,a) where $M \in Sp(2n,\mathbb{R})$ is an element of the symplectic group

$$Sp(2n,\mathbb{R}) := \left\{ M \in GL(2n,\mathbb{R}) \mid {}^t M\, J\, M = J \right\}, \qquad J = \begin{pmatrix} 0 & \mathbf{1}_n \\ -\mathbf{1}_n & 0 \end{pmatrix}, \tag{19}$$

and $a \in \mathbb{R}^{2n}$ is a $2n$-dimensional vector. This group acts on $\mathbb{R}^{2n}$ as

$$\begin{aligned} R : ISp(2n,\mathbb{R}) \times \mathbb{R}^{2n} &\longrightarrow \quad \mathbb{R}^{2n} \\ (M,a) \quad \times \ v \quad &\longmapsto R_{(M,a)} v = Mv + a\,, \end{aligned} \tag{20}$$

so that the pullback of any function $f \in \mathcal{C}^\infty(M)$ by the action of any element $(M,a) \in ISp(2n,\mathbb{R})$ reads

$$(R^*_{(M,a)} f)(y) = f(My + a)\,. \tag{21}$$

Now the Moyal–Weyl star product of the pullback of two functions by this action simply reads

$$(R^*_{(M,a)} f \star_{\text{MW}} R^*_{(M,a)} g)(y) = \exp\left(\hbar \omega^{ab} \partial_{y_1^a} \partial_{y_2^b}\right) f(My_1 + a) g(My_2 + a)|_{y_1 = y = y_2}\,, \tag{22}$$

and can be re-written as

$$(R^*_{(M,a)} f \star_{\text{MW}} R^*_{(M,a)} g)(y) = \exp\left({}^t(M\partial_{y_1}) \omega^{-1} M \partial_{y_2}\right) f(y_1) g(y_2)|_{y_1 = My + a = y_2}\,, \tag{23}$$

where the notation $M\partial_y$ should be understood as the matrix product (i.e. we think of ∂_y as a vector on which the matrix M acts). Indeed, by the chain rule, any derivative on f or g will produce a factor of M, so that a simple change of the intermediate variables y_1 and y_2 leads to the above expression. Similarly, we have written the argument of the exponential in the Moyal–Weyl product using matrix notation,

$$\partial_{y_1^a} \omega^{ab} \partial_{y_2^b} = {}^t\partial_{y_1} \omega^{-1} \partial_{y_2}\,. \tag{24}$$

Now using the fact that $M \in Sp(2n,\mathbb{R})$ preserves ω by definition, i.e.

$${}^t M \omega^{-1} M = \omega^{-1}\,, \tag{25}$$

we end up with,

$$(R^*_{(M,a)} f \star_{\text{MW}} R^*_{(M,a)} g)(y) = \left(R^*_{(M,a)} (f \star_{\text{MW}} g)\right)(y)\,, \tag{26}$$

thereby proving the equivariance of the Moyal–Weyl star-product under $ISp(2n,\mathbb{R})$.

The Gutt star-product. A simple example of Poisson manifolds is provided by the dual $\mathfrak{g}^*$ of any Lie algebra $(\mathfrak{g}, [-,-]_\mathfrak{g})$. The algebra of polynomial functions on $\mathfrak{g}^*$ naturally identifies with the symmetric algebra of $\mathfrak{g}$, i.e. $\mathcal{C}^\infty_{pol}(\mathfrak{g}^*) \cong S(\mathfrak{g})$. As a vector space, it is isomorphic to the universal enveloping algebra of $\mathfrak{g}$,

$$W : S(\mathfrak{g}) \xrightarrow{\sim} \mathcal{U}(\mathfrak{g})\,, \tag{27}$$

where W is known as the Poincaré–Birkhoff–Witt (PBW) isomorphism, and is given by

$$W(x_1 \odot \cdots \odot x_n) = \tfrac{1}{n!} \sum_{\sigma \in \mathcal{S}_n} x_{\sigma_1} \otimes \cdots \otimes x_{\sigma_n}\,, \tag{28}$$

for $x_1, \ldots, x_n \in \mathfrak{g}$. In order to lighten the expressions thereafter, we will omit the symbols $\odot$ and $\otimes$, and will simply differentiate between elements of $S(\mathfrak{g})$ and $\mathcal{U}(\mathfrak{g})$ by adding a hat on top of elements of the Lie algebra $\mathfrak{g}$ thought of as embedded its universal enveloping algebra $\mathcal{U}(\mathfrak{g})$, in contrast with elements of $\mathfrak{g}$ thought of as embedded in its symmetric algebra $S(\mathfrak{g})$, denoted without hat. For instance, if $xy \in S^2(\mathfrak{g})$, then $W(xy) = \frac{1}{2}(\hat{x}\hat{y} + \hat{y}\hat{x})$. In plain words, W sends a monomial in $S(\mathfrak{g})$ to the symmetrisation of the monomial composed of the same letters in $\mathcal{U}(\mathfrak{g})$. Its inverse is simply obtained by decomposing any given monomial on $\mathcal{U}(\mathfrak{g})$ into a sum of symmetric ones of decreasing order, upon using the commutation relations of $\mathfrak{g}$. Consider for instance the simple term $x \otimes y$ for $x, y \in \mathfrak{g}$. Writing it as the sum of its symmetric and antisymmetric part, one finds

$$\hat{x}\hat{y} = \tfrac{1}{2}(\hat{x}\hat{y} + \hat{y}\hat{x}) + \tfrac{1}{2}(\hat{x}\hat{y} - \hat{y}\hat{x}) = \tfrac{1}{2}(\hat{x}\hat{y} + \hat{y}\hat{x}) + \tfrac{1}{2}\widehat{[x,y]}_\mathfrak{g}\,, \tag{29}$$

upon using the fact that the antisymmetric part is identified with the Lie bracket in $\mathcal{U}(\mathfrak{g})$. It is now clear that the degree 2 monomial $\hat{x}\hat{y}$ is the sum of two terms in the image of W, the first one of degree 2 as well, in $S^2(\mathfrak{g})$, and the second one of degree 1, in $\mathfrak{g}$, and hence

$$W^{-1}(\hat{x}\hat{y}) = xy + \tfrac{1}{2}\,[x,y]_{\mathfrak{g}}\,. \tag{30}$$

Repeating the same steps for $\hat{x}\hat{y}\hat{z}$ with $x,y,z \in \mathfrak{g}$, one finds

$$W^{-1}(\hat{x}\hat{y}\hat{z}) = xyz + \tfrac{1}{2}\big(x[y,z]_{\mathfrak{g}} + y[x,z]_{\mathfrak{g}} + z[x,y]_{\mathfrak{g}}\big) + \tfrac{1}{6}\big([[x,y]_{\mathfrak{g}},z]_{\mathfrak{g}} + [x,[y,z]_{\mathfrak{g}}]_{\mathfrak{g}}\big)\,, \tag{31}$$

This suggests a simple way of obtaining a quantisation of the algebra of polynomial function on $\mathfrak{g}^*$, namely to consider the universal enveloping algebra and its associative structure. Note however that nowhere in the definition of the universal enveloping algebra appears the formal parameter $\hbar$, which is needed to define a star-product. In order to remedy this problem, one can consider a version of the universal enveloping algebra of $\mathfrak{g}$ over $\mathbb{R}[\![\hbar]\!]$, by which we mean

$$\mathcal{U}_\hbar(\mathfrak{g}) := \frac{T(\mathfrak{g})[\![\hbar]\!]}{I_\hbar}\,, \tag{32}$$

where $I_\hbar$ is the two-sided ideal defined by the relation

$$I_\hbar = \big\langle x\otimes y - y\otimes x - \hbar\,[x,y]_{\mathfrak{g}}\,,\ x,y\in\mathfrak{g}\big\rangle\,. \tag{33}$$

Concretely, this amounts to considering formal power series with coefficients in the universal enveloping algebra of $\mathfrak{g}$, with the modification that the Lie bracket always occurs with one power of $\hbar$. One can adapt the PBW isomorphism in this setting,

$$W : S(\mathfrak{g})[\![\hbar]\!] \xrightarrow{\sim} \mathcal{U}_\hbar(\mathfrak{g})\,, \tag{34}$$

so that, for instance

$$W(xy) = \tfrac{1}{2}\,(\hat{x}\hat{y} + \hat{y}\hat{x})\,, \qquad W^{-1}(\hat{x}\hat{y}) = xy + \tfrac{\hbar}{2}\,\{x,y\}_{\mathfrak{g}^*}\,, \tag{35}$$

where we used the fact that the Poisson bracket on linear functions on $\mathfrak{g}^*$ (which are simply elements of $\mathfrak{g}$) is the Lie bracket of $\mathfrak{g}$. One can use this isomorphism to pullback on $S(\mathfrak{g})[\![\hbar]\!]$ the product on $\mathcal{U}_\hbar(\mathfrak{g})$ [11],

$$p \star_{\text{G}} q = W^{-1}\big(W(p)W(q)\big)\,, \qquad p,q\in S(\mathfrak{g})[\![\hbar]\!]\,. \tag{36}$$

One can derive more explicit formulae, see e.g. [5, 12–14] and references therein.

The Kontsevich star-product. As a last example, let us flash the first few orders of the star-product obtained by M. Kontsevich as a consequence of the formality theorem [15]. It gives a canonical quantisation of $(\mathbb{R}^d,\pi)$ for any Poisson bivector π—in particular it is not restricted to be constant or linear as in the previous two examples. Up to order $\hbar^3$, it reads

$$\begin{aligned} f\star_{\text{K}} g = f\cdot g + \hbar\,\pi^{ab}\,\partial_a f\,\partial_b g + \hbar^2\Big(&\tfrac{1}{2}\,\pi^{ab}\pi^{cd}\,\partial_a\partial_c f\,\partial_b\partial_d g \\ &+ \tfrac{1}{3}\,\pi^{ad}\partial_d\pi^{bc}\left[\partial_a\partial_b f\,\partial_c g + \partial_c f\,\partial_a\partial_b g\right] + \tfrac{1}{6}\,\partial_d\pi^{ac}\partial_c\pi^{bd}\,\partial_a f\,\partial_b g\Big) + \mathcal{O}(\hbar^3)\,. \end{aligned} \tag{37}$$

A couple of noteworthy comments: first, one can notice that this formula is coordinates dependent, since it is given in terms of partial derivatives of f,g and the Poisson bivector π. As such, it only provides a *local* quantisation of an arbitrary Poisson manifold (M,π), meaning it defines a star-product only on an open set covered by a given coordinate system. We will see later on that this problem can be solved by patching together this local star-products, with the help of any affine connection ∇ on the manifold to be quantised. The resulting star-product is manifestly coordinate-independent, and its first few orders are given by [16]

$$\begin{aligned} f\star_\nabla g = f\cdot g + \hbar\,\pi^{ab}\,\nabla_a f\,\nabla_b g + \hbar^2\Big(&\tfrac{1}{2}\,\pi^{ab}\pi^{cd}\,\nabla_a\nabla_c f\,\nabla_b\nabla_d g \\ &+ \tfrac{1}{3}\,\pi^{ad}\nabla_d\pi^{bc}\left[\nabla_a\nabla_b f\,\nabla_c g + \nabla_c f\,\nabla_a\nabla_b g\right] + \tfrac{1}{6}\,\nabla_d\pi^{ac}\nabla_c\pi^{bd}\,\nabla_a f\,\nabla_b g\Big) + \mathcal{O}(\hbar^3)\,, \end{aligned} \tag{38}$$

which clearly agrees with the previous formula when ∇ is flat. Note that this expression looks deceiptly simple: contrarily to what this suggests, a global formula for a star-product cannot be obtained by simply replacing the partial derivatives with covariant ones in Kontsevich's (local) formula. For one thing, partial derivatives commute while covariant ones don't, hence this replacement would be ambiguous. At order $\hbar^3$ and higher, the curvature of ∇ and derivatives thereof will appear, see e.g. [16, Sec. 2] for an explicit example.

The second comment we would like to make is that the term $\frac{1}{6}\,\partial_d\pi^{ac}\partial_c\pi^{bd}\,\partial_a f\,\partial_b g$ appearing at second order in $\hbar$ can be removed without spoiling the associativity at order $\hbar^2$. In fact, one can remove it from the star-product altogether by a 'gauge transformation'. To properly explain this statement, let us introduce the notion of equivalence between star-products, and come back to this example after.

Definition 2 [**Equivalence of star-products**]**.** Two star-products $\star$ and $\star'$ are called equivalent if there exists an operator

$$T : \mathcal{C}^\infty(M)[\![\hbar]\!] \longrightarrow \mathcal{C}^\infty(M)[\![\hbar]\!]\,, \tag{39}$$

of the form $T = \mathbf{1} + \mathcal{O}(\hbar)$, and such that

$$T(f \star g) = T(f) \star' T(g)\,, \tag{40}$$

for any pair of functions $f, g \in \mathcal{C}^\infty(M)$.

Note that the fact that an operator T implementing an equivalence between two star-product start with the identity at order in $\hbar$ implies that it is invertible. Indeed, this follows from the following lemma.

Lemma 1. *A formal power series $T \in \mathrm{End}(V)[\![\hbar]\!]$ with coefficients in endomorphisms of a vector space V is invertible if and only if its coefficient at order $\hbar^0$ is.*

Proof. Assume that there exists an inverse $S = \sum_{k=0}^\infty \hbar^k\, S_k$ to an operator $T = \sum_{k=0}^\infty \hbar^k\, T_k$. Then,

$$\mathbf{1} = S \circ T = S_0 \circ T_0\,, \tag{41}$$

which implies that $S_0 = T_0^{-1}$, i.e. the coefficient of $\hbar^0$ in T is invertible.

Conversely, suppose that we are given $T = \sum_{k\geq 0} \hbar^k\, T_k$ with T_0 invertible. Then, T is invertible up to order $\hbar$ since $T_0^{-1} \circ T = \mathbf{1} + \mathcal{O}(\hbar)$. Now let us assume that we have found

$$S^{(n)} := T_0^{-1} + \sum_{k=1}^{n} \hbar^k\, S_k\,, \tag{42}$$

such that it defines an inverse of T up to order $\hbar^{n+1}$, i.e. $S^{(n)} \circ T = \mathbf{1} + \mathcal{O}(\hbar^{n+1})$, and consider $S^{(n+1)} = S^{(n)} + \hbar^{n+1}\, S_{n+1}$. Then, upon using the assumption on $S^{(n)}$, one finds

$$S^{(n+1)} \circ T = \mathbf{1} + \mathcal{O}(\hbar^{n+2}) \qquad \Leftrightarrow \qquad S_{n+1} \circ T_0 + T_0^{-1} \circ T_{n+1} + \sum_{k=1}^{n} S_k \circ T_{n+1-k} = 0\,, \tag{43}$$

i.e. $S^{(n+1)}$ defines an inverse of T up to order $\hbar^{n+2}$ if and only if S_{n+1} satisfies the above equation. Since T_0 is invertible, this condition can be used as a definition of S_{n+1},

$$S_{n+1} = -\big(T_0^{-1} \circ T_{n+1} \circ T_0^{-1} + \sum_{k=1}^{n} S_k \circ T_{n+1-k} \circ T_0^{-1}\big)\,, \tag{44}$$

which shows that an inverse of T can be constructed order by order in $\hbar$ without obstruction. ∎

In light of the previous lemma, we can re-write the fact that T defines an equivalence between the star-products $\star$ and $\star'$ as

$$f \star' g = T\left(T^{-1}(f) \star T^{-1}(g)\right) \qquad \Leftrightarrow \qquad f \star g = T^{-1}\left(T(f) \star' T(g)\right) \tag{45}$$

which makes it clear that one can be obtained as a kind of pullback of the other, thereby justifying this notion of equivalence. Said differently, an equivalence of star-products is simply an isomorphism of associative algebras,

$$T : \big(\mathcal{C}^\infty(M)[\![\hbar]\!], \star\big) \xrightarrow{\sim} \big(\mathcal{C}^\infty(M)[\![\hbar]\!], \star'\big)\,, \tag{46}$$

since T is both required to be an algebra morphism (it preserves the multiplications, which are the star-products on both sides), and is invertible.

Let us illustrate this notion with a few examples. First, we will start by revisiting the Moyal–Weyl star-product: it is apparent from the proof of its associativity that the formula makes sense for any *constant* rank-2 tensor α. The fact that ω^{ab} is antisymmetric, or even non-degenerate, does not play any rôle in ensuring its associativity. Consequently,

$$(f \star_\alpha g)(y) = \exp\big(\hbar\, \alpha^{ab} \tfrac{\partial}{\partial y_1^a} \tfrac{\partial}{\partial y_2^b}\big)\, f(y_1) g(y_2)|_{y_1=y=y_2}\,, \tag{47}$$

with

$$\alpha^{[ab]} = \omega^{ab}\,, \tag{48}$$

defines a star-product that provides a quantisation of $(\mathbb{R}^{2n}, \omega)$. As it turns out, this star-product is equivalent to the Moyal–Weyl one.

Lemma 2. *The star-product $\star_\alpha$ is equivalent to the Moyal–Weyl one.*

Proof. Let denote by $\eta^{ab} = \alpha^{(ab)}$ the symmetric part of α, so that $\alpha^{ab} = \omega^{ab} + \eta^{ab}$, and define the operator

$$T(f) = \exp\big(-\tfrac{\hbar}{2}\, \eta^{ab} \tfrac{\partial}{\partial y^a} \tfrac{\partial}{\partial y^b}\big)\, f\,, \tag{49}$$

for any $f \in \mathcal{C}^\infty(\mathbb{R}^{2n})[\![\hbar]\!]$. The claim is that this operator defines an equivalence between $\star_\alpha$ and $\star_{\mathsf{MW}}$, the Moyal–Weyl product. Indeed,

$$\begin{aligned}
\big(T(f) \star_{\mathsf{MW}} T(g)\big)(y) &= \exp\big(\hbar\omega^{ab}\, \partial_{y_1^a} \partial_{y_2^b}\big) && (50)\\
&\qquad \exp\big(-\tfrac{\hbar}{2}\, \eta^{ab} \partial_{y_1^a} \partial_{y_1^b}\big)\, \exp\big(-\tfrac{\hbar}{2}\, \eta^{ab} \partial_{y_2^a} \partial_{y_2^b}\big)\, f(y_1)\, g(y_2)|_{y_1=y=y_2} \\
&= \exp\big(-\tfrac{\hbar}{2}\, \eta^{ab} (\partial_{y_1^a} + \partial_{y_2^a})(\partial_{y_1^b} + \partial_{y_2^b})\big) && (51)\\
&\qquad \exp\big(\hbar \alpha^{ab} \partial_{y_1^a} \partial_{y_2^b}\big)\, f(y_1)\, g(y_2)|_{y_1=y=y_2}\,, \\
&= T\big(f \star_\alpha g\big)(y)\,, && (52)
\end{aligned}$$

upon using the fact that $\partial_{y_1} + \partial_{y_2}$ implements the Leibniz rule on the product $f(y_1)g(y_2)$ after the evaluation $y_1 = y = y_2$, hence the claim. ∎

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